

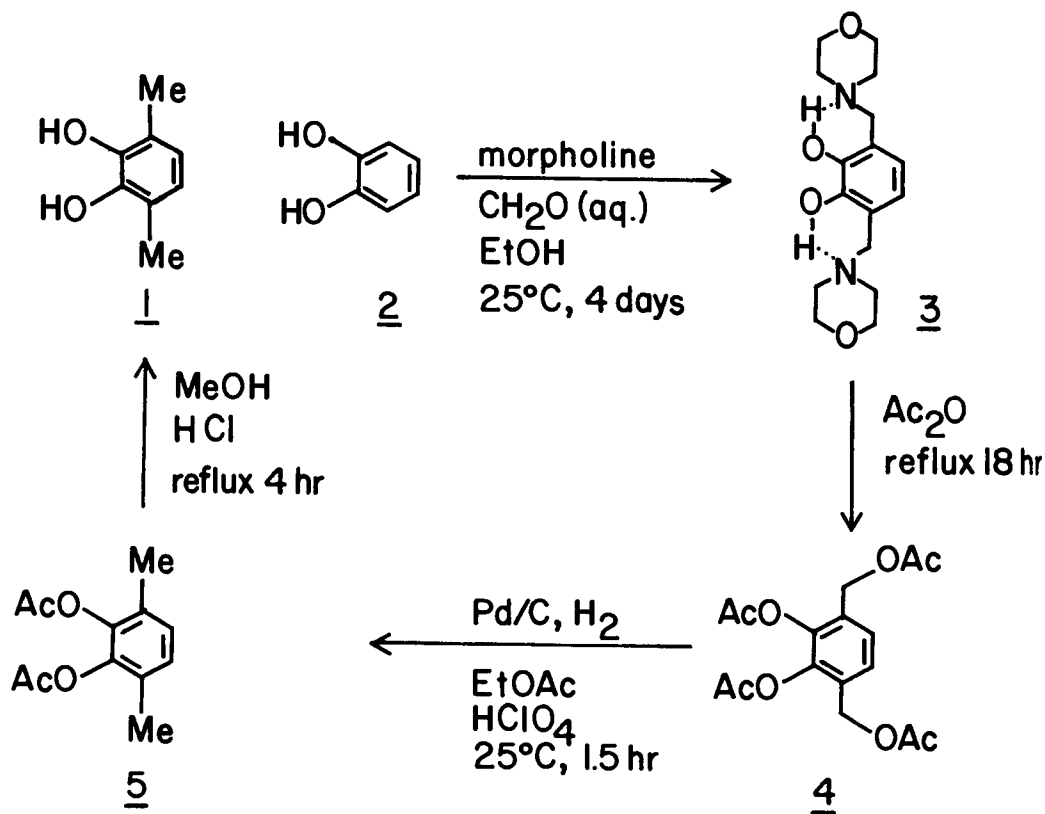
AN EFFICIENT METHOD FOR THE CONVERSION OF PHENOLIC
MANNICH BASES TO C-METHYLATED PHENOLS.
SYNTHESIS OF 3,6-DIMETHYLCATECHOL

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In connection with our synthesis of analogs of serotonin neurotoxins¹ we needed access to 3,6-dimethylcatechol (1) in quantities. This deceptively simple compound is not available commercially and is extremely inaccessible by the existing methods.²⁻⁵ One of these methods⁵ involves conversion of catechol (2) to 3,6-bis(morpholinomethyl)catechol (3) followed by hydrogenolysis using Pd/C as the catalyst. The latter step required high temperature, long reaction time and a very high catalyst (10% Pd/C) to substrate ratio (1:3 by wt.). In addition, this procedure of hydrogenolysis gave highly variable results in our hands. The difficulty encountered in the conversion of 3 to 1 is not an isolated case. Resistance of phenolic Mannich bases to direct hydrogenolysis is a general⁶⁻¹⁰ phenomenon although the exact reason for such resistance is not known and no general solution for this problem has been developed. It has been suggested⁸ that intramolecular hydrogen bonding between phenolic OH and the nitrogen atom of the amine function (cf. 3) may,

in part, be responsible for the difficulty involved in direct hydrogenolysis. However, hydrogenolysis of O-trimethylsilyl derivatives of several phenolic Mannich bases still required^{8,11} elevated temperature and long reaction time and overall yields were poor to modest. We thought a general solution to this problem would result if the amine function of the Mannich base is first suitably modified, or removed, for example, by acetolysis.¹² In this paper we report an efficient synthesis of 3,6-dimethylcatechol (1) from 3,6-bis(morpholinomethyl)catechol employing the latter approach



Catechol (2) was transformed to 3,6-bis(morpholinomethyl)catechol (3) in 75% yield by the literature procedure.⁹ Acetolysis of 3 in refluxing acetic anhydride gave the highly crystalline tetraacetate 4 in quantitative conversion. Initial attempts to hydrogenolyze

4 to 5 in ethyl acetate at 60 psi of H_2 with Pd/C as the catalyst in the absence or in the presence of acid catalysts such as HOAc or HCl were not successful. In each case, hydrogen uptake was extremely slow at room temperature. However, hydrogenolysis proceeded smoothly in the presence of small amounts of 70% perchloric acid. The reaction was complete in 1.5 hr at 25°C with initial H_2 pressure of 50 psi and a catalyst (10% Pd/C) to substrate ratio of 1:50 (by wt.). These results compare very favorably with a catalyst to substrate ratio of 1:3 (by wt.), the higher temperatures, longer reaction times and variability in the nature and yields of product for the conversion of 3 to 1 by direct hydrogenolysis. 3,6-Dimethylcatechol (1) was then produced from the diacetate 5 by transesterification with methanol in the presence of hydrochloric or perchloric acid with or without prior isolation of the diacetate 5, in 91-93% yields based on the Mannich base 3.

Since the acetolysis of phenolic Mannich bases is a widely applicable process,^{5,12} the present approach provides a convenient way for generating C-methylated phenols from Mannich bases that are resistant to direct hydrogenolysis.

EXPERIMENTAL SECTION

3,6-Bis(morpholinomethyl)catechol (3) was prepared in 75% yield from catechol (2) by the procedure of Caldwell and Thompson:⁹ mp 173-174°C (lit.⁵ mp 173-174°C; lit.⁹ mp 173-174°C); 1H NMR (acetone- d_6) δ 2.50 (t, $J=5Hz$, 8H, NCH_2CH_2O), 3.57-3.80 (m, 12H, OCH_2CH_2N and benzylic CH_2), 6.03 (br, s, 2H, OH), 6.48 (s, 2H, H-4 and H-5).

3,6-Bis(acetoxymethyl)catechol diacetate (4) was prepared in quantitative yields by the method of Fields and Reynold,⁵ by refluxing 3 in acetic anhydride (1 ml/mmol of 3) for 18 hr followed by evaporation of solvent and precipitation of crude product in water. The crude product was recrystallized twice, first from absolute ethanol after decolorizing with neutral charcoal in the same solvent, and then from ethyl acetate to give 4 as a white granular solid (for smooth hydrogenolysis of the tetraacetate 4 it was essential to purify the crude product in this manner): mp 114-115°C (lit.⁵ mp 113-114°C); ¹H NMR (CDCl₃) δ 2.05 (s, 6H, benzylic C(O)CH₃), 2.30 (s, 6H, phenolic C(O)CH₃), 5.07 (s, 4H, ArCH₂), 7.37 (s, 2H, H-4 and H-6).

3,6-Dimethylcatechol. Method A: Without Prior Isolation of Diacetate 4. The tetraacetate 4 (20.1 g, 59.4 mmol) was dissolved in 280 ml of ethyl acetate by warming. The solution was placed in a 500 ml Parr hydrogenation bottle and then were added, in order, 2 ml of 70% HClO₄ and a suspension of 10% Pd/C (400 mg) in 20 ml of ethyl acetate. The mixture was hydrogenated at 50 psi (initial pressure) at room temperature. After theoretical pressure drop had occurred (about 1 to 1.5 hr), hydrogenation was terminated, the catalyst was allowed to settle (1 hr), and was removed by gravity filtration. The filtrate was then concentrated in vacuo to remove ethyl acetate. The concentrate was dissolved in methanol (100 ml), 1 ml 70% HClO₄ was added and the resulting mixture was refluxed for 4 hr under nitrogen. After cooling, the mixture was neutralized to pH 7-8 with solid NaHCO₃ and then filtered. The filtrate was evaporated

in vacuo to dryness. The solid residue thus produced was extracted with boiling hexane (3 x 100 ml). Evaporation of the combined hexane extracts to dryness, followed by recrystallization of the crude solid from hexane gave 1 as a white amorphous solid (7.62 g, 93%): mp 99-101°C (lit.⁵ mp 98.5-99.5°C, lit.⁹ mp 101°C); ¹H NMR (CDCl₃) δ 2.23 (s, 6H, Me), 5.07 (br, s, 2H, OH), 6.63 (s, 2H, H-4 and H-5).

Method B: With Isolation of Diacetate 5. Hydrogenation was carried out as described in Method A. After removal of the catalyst, the filtrate was washed with water (2 x 150 ml), satd. NaHCO₃ (1 x 50 ml or until all acidic materials were removed from the organic layers), satd. NaCl (1 x 50 ml), dried (Na₂SO₄) and then evaporated in vacuo to dryness to give 12.4 g (94%) of 4 as a white granular solid: mp 126-130°C. One recrystallization from acetone-hexane raised the mp to 130-132°C: ¹H NMR (CDCl₃) δ 2.17 (s, 6H, Me), 2.30 (s, 6H, acetyl), 7.03 (s, 2H, H-4 and H-5).

The diacetate 5 (12.4 g) was dissolved in 100 ml of methanol and 2 ml of conc. HCl was added. The mixture was refluxed for 4 hr under N₂ and then cooled and neutralized to pH 7-8 with solid NH₄HCO₃. The mixture was filtered and the filtrate was evaporated in vacuo to dryness. The solid residue was extracted with boiling hexane (3 x 100 ml). The combined hexane extracts gave 7.45 g of 1 (97% based on the diacetate 5 and 91% based on the tetraacetate 4) which was identical in all respects with 1 obtained by Method A.

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